



SIMATS Online Programming Lab
JAVA PROGRAMMING

SANGEPU VENKATA PRAVEEN
192373013RD

QUESTIONS 20 / 20 completed

Bank

BANK
Write a Java program to implement a basic banking system using inheritance. Create a parent class called Account with attributes such as account number, account holder name, and balance. Then, create child classes such as

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Account {
3     protected String accountNumber;
4     protected String accountHolderName;
5     protected double balance;
6     Account(String accountNumber, String accountHolderName, double balance) {
7         this.accountNumber = accountNumber;
8         this.accountHolderName = accountHolderName;
9         this.balance = balance;
10    }
11    void deposit(double amount) {
12        balance += amount;
13    }
14    double getBalance() {
15        return balance;
16    }
17    public static void main(String[] args) {
18        Scanner sc = new Scanner(System.in);
19        SavingsAccount savings =
20            new SavingsAccount("123456789", "Jane Doe", 5000, 1.5);
21    }
```

INPUT

```
1
deposit
2500
2
withdraw
2000
```



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account number,
account holder name,
and balance. Then,
create child classes
such as

```
18 Scanner sc = new Scanner(System.in);
19 SavingsAccount savings =
20     new SavingsAccount("123456789", "Jane Doe", 5000, 1.5);
21 CheckingAccount checking =
22     new CheckingAccount("987654321", "John Smith", 1000, 1000);
23 System.out.println("\nThe accounts in the system are:\n");
24 System.out.println("1. Savings Account (" + savings.accountNumber + ") - " + s
25     " (Balance: $" + String.format("%.2f", savings.getBalance()) + ", Inte
26     savings.getInterestRate() + "%)");
```

OUTPUT

```
The accounts in the system are:

1. Savings Account (123456789) - Jane Doe (Balance: $5000.00, Interest Rate:
1.5%)
2. Checking Account (987654321) - John Smith (Balance: $1000.00, Overdraft
Limit: $1000.00)
Please enter the transaction type (deposit/withdraw): Please enter the amount
to deposit:
Transaction successful. New balance: $7500.00
```



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QUESTIONS 20 / 20 completed

Employee

EMPLOYEE

Write a Java program to implement a basic employee management system using inheritance. Create a parent class called Employee with attributes such as name, id, and salary. Then, create child classes such as Manager and

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Employee {
3     String name;
4     int id;
5     double salary;
6     Employee(String name, int id, double salary) {
7         this.name = name;
8         this.id = id;
9         this.salary = salary;
10    }
11    void display() {
12        System.out.println("Name: " + name);
13        System.out.println("ID: " + id);
14        System.out.printf("Salary: $%.2f\n", salary);
15    }
16    public static void main(String[] args) {
17        Scanner sc = new Scanner(System.in);
18        String managerName = sc.nextLine();
19        int managerId = sc.nextInt();
20        double managerSalary = sc.nextDouble();
21        int teamMembers = sc.nextInt();
```

INPUT

Alice
101
75000.50
3
Bob
102
50000.00
12500.75



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attributes such as name, id, and salary. Then, create child classes such as Manager and

```
25     String salesName = sc.nextLine();
26     int salesId = sc.nextInt();
27     double salesSalary = sc.nextDouble();
28     double salesAmount = sc.nextDouble();
29     SalesPerson salesPerson = new SalesPerson(
30         salesName, salesId, salesSalary);
31     System.out.println("\n--- Manager Details ---");
32     manager.display();
33     manager.manageTeam();
34     System.out.println("\n--- SalesPerson Details ---");
```

OUTPUT

--- Manager Details ---
Name: Alice
ID: 101
Salary: \$75000.50
Managing a team of 3 members.

--- SalesPerson Details ---
Name: Bob
ID: 102
Salary: \$50000.00



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QUESTIONS 20 / 20 completed

Animal-1

ANIMAL-1

Write a Java program to implement a basic animal classification system using inheritance. Create a parent class called Animal with attributes such as name and habitat. Then, create child classes such as Mammal, Reptile, and Bird that inherit from

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Animal {
3     protected String name;
4     protected String habitat;
5     public Animal(String name, String habitat) {
6         this.name = name;
7         this.habitat = habitat;
8     }
9     public void displayInfo() {
10        System.out.println("Name: " + name + ", Habitat: " + habitat);
11    }
12    public void reproduce() {
13        System.out.println("Animal reproduces.");
14    }
15    public static void main(String[] args) {
16        Mammal lion = new Mammal("Lion", "Savannah");
17        Reptile snake = new Reptile("Snake", "Desert");
18        Bird eagle = new Bird("Eagle", "Mountains");
19        System.out.println("--- Mammal ---");
20        lion.displayInfo();
21        lion.eat();
22    }
```

INPUT

Your INPUT goes here!



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parent class called Animal with attributes such as name and habitat. Then, create child classes such as Mammal, Reptile, and Bird that inherit from

```
14 }
15 public static void main(String[] args) {
16     Mammal lion = new Mammal("Lion", "Savannah");
17     Reptile snake = new Reptile("Snake", "Desert");
18     Bird eagle = new Bird("Eagle", "Mountains");
19     System.out.println("--- Mammal ---");
20     lion.displayInfo();
21     lion.eat();
22     lion.reproduce();
23     System.out.println("\n--- Reptile ---");
24     snake.displayInfo();
25 }
```

OUTPUT

```
Name: Lion, Habitat: Savannah
Lion eats milk/plants.
Lion gives birth to live young.

--- Reptile ---
Name: Snake, Habitat: Desert
Snake eats insects/meat.
Snake lays eggs.

--- Bird ---
```



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QUESTIONS 20 / 20 completed

Shapes-1

SHAPES-1

Write a Java program to implement a basic shape hierarchy using inheritance. Create a parent class called Shape with attributes such as color and area. Then, create child classes such as Circle, Square, and Triangle that inherit from the Shape class and have

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Circle circle = new Circle("Red", 5.0);
5         circle.calculateArea();
6         circle.displayInfo();
7         circle.calculatePerimeter();
8         System.out.println();
9         Square square = new Square("Blue", 4.0);
10        square.calculateArea();
11        square.displayInfo();
12        square.calculatePerimeter();
13        System.out.println();
14        Triangle triangle = new Triangle("Green", 3.0, 4.0, 3.0, 4.0, 5.0);
15        triangle.calculateArea();
16        triangle.displayInfo();
17        triangle.calculatePerimeter();
18    }
19 }
20 class Shape {
21
```

INPUT

Your INPUT goes here!



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Then, create child classes such as Circle, Square, and Triangle that inherit from the Shape class and have

```
17        triangle.calculatePerimeter();
18    }
19 }
20 class Shape {
21     String color;
22     double area;
23     Shape(String color) {
24         this.color = color;
25         this.area = 0.0;
26
```

OUTPUT

Color: Red, Area: 78.53981633974483
Perimeter (Circumference): 31.41592653589793

Color: Blue, Area: 16.0
Perimeter: 16.0

Color: Green, Area: 6.0
Perimeter: 12.0



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QUESTIONS 20 / 20 completed

Library

LIBRARY

Write a Java program to implement a basic library management system using inheritance. Create a parent class called Item with attributes such as title and author. Then, create child classes such as Book and DVD that inherit from the Item

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Book myBook = new Book("The Great Gatsby", "F. Scott Fitzgerald", "ISBN-12345");
5         myBook.displayDetails();
6         myBook.borrowItem();
7         myBook.returnItem();
8         System.out.println("-----");
9         DVD myDVD = new DVD("Inception", "Christopher Nolan", "90 Minutes");
10        myDVD.displayDetails();
11        myDVD.borrowItem();
12        myDVD.returnItem();
13    }
14 }
15 class Item {
16     protected String title;
17     protected String author;
18     protected boolean isBorrowed;
19     public Item(String title, String author) {
20         this.title = title;
21     }
22 }
```

INPUT

Your INPUT goes here!



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parent class called Item with attributes such as title and author. Then, create child classes such as Book and DVD that inherit from the Item

```
10        myDVD.displayDetails();
11        myDVD.borrowItem();
12        myDVD.returnItem();
13    }
14 }
15 class Item {
16     protected String title;
17     protected String author;
18     protected boolean isBorrowed;
19     public Item(String title, String author) {
20         this.title = title;
21     }
22 }
```

OUTPUT

```
Title: The Great Gatsby
Author: F. Scott Fitzgerald
Type: Book
ISBN: ISBN-12345
Successfully borrowed: The Great Gatsby
Successfully returned: The Great Gatsby
-----
Title: Inception
Author: Christopher Nolan
Type: DVD
```



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QUESTIONS 20 / 20 completed

University

UNIVERSITY

Write a Java program to implement a basic university management system using inheritance. Create a parent class called Person with attributes such as name and age. Then, create child classes such as Student and Professor that inherit

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Student student = new Student("Alice", 20, "S12345", "Computer Science");
5         Professor professor = new Professor("Dr. Smith", 45, "P98765", "Java Programmi
6         System.out.println("--- Student Information ---");
7         student.displayStudentDetails();
8         System.out.println("\n--- Professor Information ---");
9         professor.displayProfessorDetails();
10    }
11 }
12 class Person {
13     String name;
14     int age;
15     Person(String name, int age) {
16         this.name = name;
17         this.age = age;
18     }
19     void displayPersonInfo() {
20         System.out.println("Name: " + name);
```

INPUT

Your INPUT goes here!



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called Person with attributes such as name and age. Then, create child classes such as Student and Professor that inherit

```
11 }
12 class Person {
13     String name;
14     int age;
15     Person(String name, int age) {
16         this.name = name;
17         this.age = age;
18     }
19     void displayPersonInfo() {
20         System.out.println("Name: " + name);
21 }
```

OUTPUT

```
--- Student Information ---
Name: Alice
Age: 20
Student ID: S12345
Alice is studying Computer Science.

--- Professor Information ---
Name: Dr. Smith
Age: 45
Employee ID: P98765
```



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QUESTIONS 20 / 20 completed

Games

GAMES

Write a Java program to implement a basic game system using inheritance. Create a parent class called Game with attributes such as title and genre. Then, create child classes such as ActionGame and PuzzleGame that inherit from the Game

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Game {
3     String title;
4     String genre;
5     Game(String title, String genre) {
6         this.title = title;
7         this.genre = genre;
8     }
9     void displayDetails() {
10        System.out.println("Title: " + title + ", Genre: " + genre);
11    }
12    void play() {
13        System.out.println("Playing the game...");
14    }
15    public static void main(String[] args) {
16        Game g1 = new ActionGame("Shadow Run", "Action", 5, 3);
17        Game g2 = new PuzzleGame("Mind Maze", "Puzzle", 10, "Logic");
18        g1.displayDetails();
19        ((ActionGame) g1).displayActionDetails();
20        g1.play();
21        System.out.println();
```

INPUT

Your INPUT goes here!



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child classes such as ActionGame and PuzzleGame that inherit from the Game

```
48     this.maxMoves = maxMoves;
49     this.logicType = logicType;
50 }
51 void displayPuzzleDetails() {
52     System.out.println("Max Moves: " + maxMoves + ", Logic Type: " + logicType);
53 }
54 @Override
55 void play() {
```

OUTPUT

```
Title: Shadow Run, Genre: Action
Difficulty: 5, Level: 3
Playing an intense action game!

Title: Mind Maze, Genre: Puzzle
Max Moves: 10, Logic Type: Logic
Solving a challenging puzzle!
```



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QUESTIONS 20 / 20 completed

Resturant

RESTURANT

Write a Java program to implement a basic restaurant management system using inheritance. Create a parent class called Menu with attributes such as name and price. Then, create child classes such as Appetizer and Entree that inherit

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Appetizer app = new Appetizer("Spring Rolls", 5.99, "Savory");
5         app.displayDetails();
6         app.orderAppetizer();
7         app.serveAppetizer();
8         System.out.println();
9         Entree ent = new Entree("Grilled Salmon", 18.50, "Salmon");
10        ent.displayDetails();
11        ent.orderEntree();
12        ent.serveEntree();
13    }
14 }
15 class Menu {
16     String name;
17     double price;
18     Menu(String name, double price) {
19         this.name = name;
20         this.price = price;
21     }
22 }
```

INPUT

Your INPUT goes here!



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name and price. Then, create child classes such as Appetizer and Entree that inherit

```
30     flavor = flavor;
31 }
32 void orderAppetizer() {
33     System.out.println("Ordering appetizer: " + name + " (" + flavor + ")");
34 }
35 void serveAppetizer() {
36     System.out.println("Serving " + name + " as a starter!");
37 }
38 }
```

OUTPUT

Item: Spring Rolls | Price: \$5.99
Ordering appetizer: Spring Rolls (Savory)
Serving Spring Rolls as a starter!

Item: Grilled Salmon | Price: \$18.5
Ordering entree: Grilled Salmon with Salmon
Serving Grilled Salmon as the main course!



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QUESTIONS 20 / 20 completed

Forecasting

FORCASTING
Write a Java program to implement a basic weather forecasting system using inheritance. Create a parent class called Forecast with attributes such as date and temperature. Then, create child classes such as SunnyForecast and

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String date = sc.nextLine();
7         double temperature = sc.nextDouble();
8         String type = sc.next();
9         if (type.equalsIgnoreCase("sunny")) {
10             System.out.println("Enter UV Index:");
11             int uvIndex = sc.nextInt();
12             SunnyForecast sunny = new SunnyForecast(date, temperature, uvIndex);
13             sunny.displayForecast();
14             sunny.predictWeather();
15         } else if (type.equalsIgnoreCase("rainy")) {
16             System.out.println("Enter Rainfall Amount:");
17             double rainfall = sc.nextDouble();
18             RainyForecast rainy = new RainyForecast(date, temperature, rainfall);
19             rainy.displayForecast();
20             rainy.predictWeather();
21         } else {
```

INPUT

2024-06-15
32.5
sunny
8



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parent class called Forecast with attributes such as date and temperature. Then, create child classes such as SunnyForecast and

PROGRAM EDITOR

Java

Run Save

```
10 System.out.println("Enter UV Index:");
11 int uvIndex = sc.nextInt();
12 SunnyForecast sunny = new SunnyForecast(date, temperature, uvIndex);
13 sunny.displayForecast();
14 sunny.predictWeather();
15 } else if (type.equalsIgnoreCase("rainy")) {
16 System.out.println("Enter Rainfall Amount:");
17 double rainfall = sc.nextDouble();
18 RainyForecast rainy = new RainyForecast(date, temperature, rainfall);
19 rainy.displayForecast();
20 rainy.predictWeather();
21 } else {
```

OUTPUT

Enter UV Index:
Date: 2024-06-15
Temperature: 32.5 C
Weather Type: Sunny
UV Index: 8
Advice: Use sunscreen, high UV levels.

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QUESTIONS 20 / 20 completed

Rental

RENTAL

Write a Java program to implement a basic car rental system using inheritance. Create a parent class called Vehicle with attributes such as make and model. Then, create child classes such as Sedan and SUV that inherit from the Vehicle class and have

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Sedan sedan = new Sedan("Toyota", "Camry", 4);
4         SUV suv = new SUV("Ford", "Explorer", 7);
5         System.out.println("--- Sedan Details ---");
6         sedan.displaySedanInfo();
7         sedan.rent();
8         sedan.rent();
9         sedan.returnVehicle();
10        System.out.println("\n--- SUV Details ---");
11        suv.displaySUVInfo();
12        suv.rent();
13        suv.returnVehicle();
14    }
15 }
16 class Vehicle {
17     String make;
18     String model;
19     boolean isRented;
20     Vehicle(String make, String model) {
21         this.make = make;
```

INPUT

Your INPUT goes here!

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model. Then, create child classes such as Sedan and SUV that inherit from the Vehicle class and have

```
12        suv.rent();
13        suv.returnVehicle();
14    }
15 }
16 class Vehicle {
17     String make;
18     String model;
19     boolean isRented;
20     Vehicle(String make, String model) {
21         this.make = make;
```

OUTPUT

```
--- Sedan Details ---
Make: Toyota, Model: Camry
Number of Doors: 4
Toyota Camry has been rented.
Toyota Camry is already rented.
Toyota Camry has been returned.

--- SUV Details ---
Make: Ford, Model: Explorer
Seating Capacity: 7
```



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QUESTIONS 20 / 20 completed

Shape-2

SHAPE-2

Write a Java program to implement a basic shape hierarchy using polymorphism. Create a parent class called Shape with attributes such as area and perimeter. Then, create child classes such as Circle and Rectangle that inherit from the Shape class

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Shape circle = new Circle(5);
5         Shape rectangle = new Rectangle(4, 6);
6         circle.display();
7         rectangle.display();
8     }
9 }
10 class Shape {
11     double area;
12     double perimeter;
13     void display() {
14         System.out.println("Area: " + area);
15         System.out.println("Perimeter: " + perimeter);
16     }
17 }
18 class Circle extends Shape {
19     double radius;
20     Circle(double radius) {
21         this.radius = radius;
```

INPUT

Your INPUT goes here!



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perimeter. Then, create child classes such as Circle and Rectangle that inherit from the Shape class

```
26     System.out.println("Circle Area: " + area);
27     System.out.println("Circle Perimeter: " + perimeter);
28 }
29 }
30 class Rectangle extends Shape {
31     double length;
32     double width;
33     Rectangle(double length, double width) {
34         this.length = length;
35         this.width = width;
```

OUTPUT

```
Area: 78.53981633974483
Perimeter: 31.41592653589793
Area: 24.0
Perimeter: 20.0
```



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QUESTIONS 20 / 20 completed

Draw

DRAW

Write a Java program to demonstrate polymorphism with interfaces. Create an interface called Drawable with a method called draw(). Then, create classes such as Circle and Square that implement the Drawable interface and have their own

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Drawable shape1 = new Circle();
4         Drawable shape2 = new Square();
5         shape1.draw();
6         shape2.draw();
7     }
8 }
9 interface Drawable {
10     void draw();
11 }
12 class Circle implements Drawable {
13     @Override
14     public void draw() {
15         System.out.println("Drawing a Circle");
16     }
17 }
18 class Square implements Drawable {
19     @Override
20     public void draw() {
21         System.out.println("Drawing a Square");
22     }
23 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 20 / 20 completed

Draw

DRAW

Write a Java program to demonstrate polymorphism with interfaces. Create an interface called Drawable with a method called draw(). Then, create classes such as Circle and Square that implement the Drawable interface and have their own

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Drawable shape1 = new Circle();
4         Drawable shape2 = new Square();
5         shape1.draw();
6         shape2.draw();
7     }
8 }
9 interface Drawable {
10     void draw();
11 }
12 class Circle implements Drawable {
13     @Override
14     public void draw() {
15         System.out.println("Drawing a Circle");
16     }
17 }
18 class Square implements Drawable {
19     @Override
20     public void draw() {
21         System.out.println("Drawing a Square");
22     }
23 }
```

OUTPUT

Drawing a Circle
Drawing a Square



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QUESTIONS 20 / 20 completed

Animal-2

ANIMAL-2
Write a Java program to demonstrate polymorphism with abstract classes. Create an abstract class called Animal with abstract methods such as eat() and sleep(). Then, create child classes such as Dog and Cat that inherit from the

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     static int sum(int... numbers) {
3         int total = 0;
4         for (int num : numbers) {
5             total += num;
6         }
7         return total;
8     }
9     static double sum(double... numbers) {
10        double total = 0;
11        for (double num : numbers) {
12            total += num;
13        }
14        return total;
15    }
16    public static void main(String[] args) {
17        int intResult = sum(1, 2, 3, 4, 5);
18        System.out.println("Sum of integers: " + intResult);
19        double doubleResult = sum(1.5, 2.5, 3.5);
20        System.out.println("Sum of doubles: " + doubleResult);
21    }
```

INPUT

Your INPUT goes here!



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such as eat() and sleep(). Then, create child classes such as Dog and Cat that inherit from the

12 total += num;

13 }

14 return total;

15 }

16 public static void main(String[] args) {

17 int intResult = sum(1, 2, 3, 4, 5);

18 System.out.println("Sum of integers: " + intResult);

19 double doubleResult = sum(1.5, 2.5, 3.5);

20 System.out.println("Sum of doubles: " + doubleResult);

21 }

OUTPUT

Sum of integers: 15
Sum of doubles: 7.5



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QUESTIONS 20 / 20 completed

Overloading

OVERLOADING

Write a Java program to demonstrate method overloading with variable arguments. Create a method called sum() that takes in a variable number of integers and returns their sum. Then, create an overloaded version of the sum() method that

PROGRAM EDITOR

Java

Run Save

```
1 public class Main {
2     public static int sum(int... numbers) {
3         int total = 0;
4         for (int n : numbers) {
5             total += n;
6         }
7         return total;
8     }
9     public static double sum(double... numbers) {
10        double total = 0.0;
11        for (double n : numbers) {
12            total += n;
13        }
14        return total;
15    }
16    public static void main(String[] args) {
17        System.out.println("Sum of integers: " + sum(1, 2, 3, 4, 5));
18        System.out.println("Sum of doubles: " + sum(1.5, 2.5, 3.0));
19    }
20 }
```

INPUT

Your INPUT goes here!



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method overloading with variable arguments. Create a method called sum() that takes in a variable number of integers and returns their sum. Then, create an overloaded version of the sum() method that

```
7     return total;
8 }
9 public static double sum(double... numbers) {
10    double total = 0.0;
11    for (double n : numbers) {
12        total += n;
13    }
14    return total;
15 }
16 public static void main(String[] args) {
17    System.out.println("Sum of integers: " + sum(1, 2, 3, 4, 5));
18    System.out.println("Sum of doubles: " + sum(1.5, 2.5, 3.0));
19 }
20 }
```

OUTPUT

Sum of integers: 15
Sum of doubles: 7.0



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QUESTIONS 20 / 20 completed

Overriding

OVERRIDING

Write a Java program to demonstrate method overriding with a simple calculator. Create a parent class called Calculator with methods such as add(), subtract(), multiply(), and divide(). Then, create a child class called

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Calculator basicCalc = new Calculator();
4         ScientificCalculator sciCalc = new ScientificCalculator();
5         System.out.println("--- Basic Calculator ---");
6         basicCalc.add(10, 5);
7         basicCalc.subtract(10, 5);
8         basicCalc.multiply(10, 5);
9         basicCalc.divide(10, 5);
10        System.out.println("\n--- Scientific Calculator ---");
11        sciCalc.add(10, 5);
12        sciCalc.subtract(10, 5);
13        sciCalc.multiply(10, 5);
14        sciCalc.divide(10, 5);
15    }
16 }
17 class Calculator {
18     public void add(double a, double b) {
19         System.out.println("Addition: " + (a + b));
20     }
21 }
```

INPUT

Your INPUT goes here!



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methods such as add(), subtract(), multiply(), and divide(). Then, create a child class called

```
12        sciCalc.subtract(10, 5);
13        sciCalc.multiply(10, 5);
14        sciCalc.divide(10, 5);
15    }
16 }
17 class Calculator {
18     public void add(double a, double b) {
19         System.out.println("Addition: " + (a + b));
20     }
21 }
```

OUTPUT

--- Basic Calculator ---
Addition: 15.0
Subtraction: 5.0
Multiplication: 50.0
Division: 2.0

--- Scientific Calculator ---
Addition: 15.0
Subtraction: 5.0
Scientific Multiplication (Result Squared): 2500.0



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QUESTIONS 20 / 20 completed

Person

PERSON

Write a Java program to create a class called Person with a constructor that takes in a name and age, and a method to print out the person's name and age.

PROGRAM EDITOR

Java

Run

Save

```
1 class Person {
2     String name;
3     int age;
4     Person(String name, int age) {
5         this.name = name;
6         this.age = age;
7     }
8     void displayInfo() {
9         System.out.println("Name: " + name + ", Age: " + age);
10    }
11    public static void main(String[] args) {
12        Person person = new Person("Venkata Praveen", 20);
13        person.displayInfo();
14    }
15 }
```

INPUT

Your INPUT goes here!



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```
14     }
15 }
```

OUTPUT

Name: Venkata Praveen, Age: 20

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QUESTIONS 20 / 20 completed

Rect-Const

RECT-CONST

Write a Java program to create a class called Rectangle with a constructor that takes in the length and width of the rectangle, and a method to calculate and return the area of the rectangle.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Rectangle rect = new Rectangle(5.0, 10.0);
4         System.out.println(rect.calculateArea());
5     }
6 }
7 class Rectangle {
8     double length;
9     double width;
10    Rectangle(double length, double width) {
11        this.length = length;
12        this.width = width;
13    }
14    double calculateArea() {
15        return length * width;
16    }
17 }
```

INPUT

Your INPUT goes here!

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```
14    double calculateArea() {
15        return length * width;
16    }
17 }
```

OUTPUT

50.0



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QUESTIONS 20 / 20 completed

Bank-Const

BANK-CONST

Write a Java program to create a class called BankAccount with a constructor that takes in an account number and an initial balance, and methods to deposit and withdraw money from the account.

PROGRAM EDITOR

Java

Run

Save

```
1 class BankAccount {
2     private String accountNumber;
3     private double balance;
4     public BankAccount(String accountNumber, double initialBalance) {
5         this.accountNumber = accountNumber;
6         this.balance = initialBalance;
7     }
8     public void deposit(double amount) {
9         if (amount > 0) {
10             balance += amount;
11             System.out.println("Successfully deposited: " + amount);
12         } else {
13             System.out.println("Invalid deposit amount.");
14         }
15     }
16     public void withdraw(double amount) {
17         if (amount > 0 && amount <= balance) {
18             balance -= amount;
19             System.out.println("Successfully withdrawn: " + amount);
20         } else {
21             System.out.println("Invalid withdrawal amount or insufficient funds.");
22         }
23     }
24 }
```

INPUT

Your INPUT goes here!



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```
14     }
15 }
16 public void withdraw(double amount) {
17     if (amount > 0 && amount <= balance) {
18         balance -= amount;
19         System.out.println("Successfully withdrawn: " + amount);
20     } else {
21         System.out.println("Invalid withdrawal amount or insufficient funds.");
22     }
23 }
```

OUTPUT

```
Account: 123456789 | Balance: 1000.0
Successfully deposited: 500.0
Successfully withdrawn: 200.0
Account: 123456789 | Balance: 1300.0
```

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QUESTIONS 20 / 20 completed

Car-Const

CAR-CONST

Write a Java program to create a class called Car with a constructor that takes in the make, model, and year of the car, and a method to print out the car's make, model, and year.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     static class Car {
3         private String make;
4         private String model;
5         private int year;
6         public Car(String make, String model, int year) {
7             this.make = make;
8             this.model = model;
9             this.year = year;
10        }
11        public void printCarDetails() {
12            System.out.println("Make: " + make);
13            System.out.println("Model: " + model);
14            System.out.println("Year: " + year);
15        }
16    }
17    public static void main(String[] args) {
18        Car c = new Car("Toyota", "Corolla", 2020);
19        c.printCarDetails();
20    }
21 }
```

INPUT

Your INPUT goes here!

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14 System.out.println("Year: " + year);
15 }
16 }
17 public static void main(String[] args) {
18 Car c = new Car("Toyota", "Corolla", 2020);
19 c.printCarDetails();
20 }
21 }

OUTPUT

Make: Toyota
Model: Corolla
Year: 2020

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QUESTIONS 20 / 20 completed

Student-Const

STUDENT-CONST

Write a Java program to create a class called Student with a constructor that takes in a name, ID number, and a list of grades, and methods to calculate and return the student's average grade and letter grade.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.List;
2 import java.util.Arrays;
3 class Student {
4     private String name;
5     private String idNumber;
6     private List<Double> grades;
7     public Student(String name, String idNumber, List<Double> grades) {
8         this.name = name;
9         this.idNumber = idNumber;
10        this.grades = grades;
11    }
12    public double calculateAverage() {
13        if (grades == null || grades.isEmpty()) {
14            return 0.0;
15        }
16        double sum = 0;
17        for (double grade : grades) {
18            sum += grade;
19        }
20        return sum / grades.size();
21    }
22 }
```

INPUT

Your INPUT goes here!

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grade.

```
14         return 0.0;
15     }
16     double sum = 0;
17     for (double grade : grades) {
18         sum += grade;
19     }
20     return sum / grades.size();
21 }
```

OUTPUT

Average Grade: 86.25
Letter Grade: B